

**This assignment will not be graded and is only for practice.**

**Multiple Select Questions (MSQ):**

1) Match the systems of linear equations in Column A with their number of solutions in **1 point** Column B and their geometric representation in Column C.

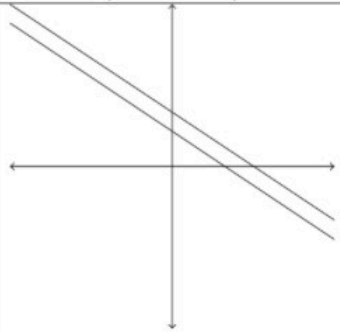
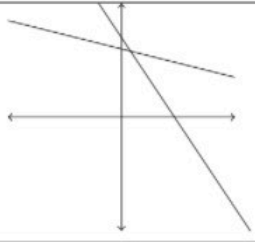
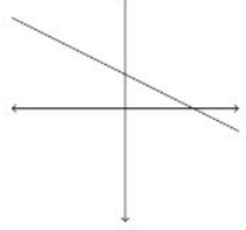
|      | System of linear equations<br>(Column A) |    | Number of solutions<br>(Column B) |    | Geometric representations<br>(Column C)                                               |
|------|------------------------------------------|----|-----------------------------------|----|---------------------------------------------------------------------------------------|
| i)   | $3x + 2y = 7, x + 4y = 12$               | a) | Infinite solutions                | 1) |    |
| ii)  | $2x + 3y = 5, 8x + 12y = 13$             | b) | No solution                       | 2) |   |
| iii) | $x + 2y = 3, 4x + 8y = 12$               | c) | Unique solution                   | 3) |  |

Table: M2W1PT1

- ☐ i) → c) → 2)
- ☐ i) → a) → 1)
- ☐ ii) → c) → 3)
- ☐ iii) → a) → 2)
- ☐ ii) → b) → 1)

No, the answer is incorrect.

Score: 0

Accepted Answers:

i)  $\rightarrow$  c)  $\rightarrow$  2)ii)  $\rightarrow$  b)  $\rightarrow$  1)

2) Choose the set of correct options

**1 point**

- ☐ If both  $A$  and  $B$  are  $2 \times 2$  real matrices and  $\det(AB) = 0$ , then  $\det(A) = 0$  or  $\det(B) = 0$ .
- ☐ If  $A$  is a  $3 \times 3$  real matrix with non-zero determinant and  $k$  is some real number, then  $\det(kA) = k^3 \times \det(A)$ .
- ☐ If  $A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$ , then  $A^{10} = 2^{10} \begin{bmatrix} 1 & 10 \\ 0 & 1 \end{bmatrix}$  (where  $A^n$  is the matrix  $A \times A \times \dots \times A$ ,  $n$ -times).
- ☐ The number of scalar additions to be done to compute the matrix  $AB$ , where  $A$  is a  $3 \times 2$  matrix and  $B$  is a  $2 \times 3$  matrix, is 9.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If both  $A$  and  $B$  are  $2 \times 2$  real matrices and  $\det(AB) = 0$ , then  $\det(A) = 0$  or  $\det(B) = 0$ .If  $A$  is a  $3 \times 3$  real matrix with non-zero determinant and  $k$  is some real number, then  $\det(kA) = k^3 \times \det(A)$ .If  $A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$ , then  $A^{10} = 2^{10} \begin{bmatrix} 1 & 10 \\ 0 & 1 \end{bmatrix}$  (where  $A^n$  is the matrix  $A \times A \times \dots \times A$ ,  $n$ -times).The number of scalar additions to be done to compute the matrix  $AB$ , where  $A$  is a  $3 \times 2$  matrix and  $B$  is a  $2 \times 3$  matrix, is 9.

3) Choose the set of correct options

**1 point**

- ☐ A triangular  $3 \times 3$  matrix has non-zero determinant if and only if all the diagonal entries are non-zero.
- ☐ If  $A$  and  $B$  are  $3 \times 3$  matrices then  $\det(A + B) = \det(A) + \det(B)$ .
- ☐ If  $A$  is a  $3 \times 3$  matrix and  $B$  is a matrix obtained from  $A$  by multiplying each column of  $A$  by its column number, then  $\det(B) = 6\det(A)$ .

- ☐ If the sum of the first and the third row vectors of a  $3 \times 3$  matrix  $A$  is equal to the second row vector of  $A$ , then  $\det(A) = 0$ .

No, the answer is incorrect.

Score: 0

### Accepted Answers:

A triangular  $3 \times 3$  matrix has non-zero determinant if and only if all the diagonal entries are non-zero.

If  $A$  is a  $3 \times 3$  matrix and  $B$  is a matrix obtained from  $A$  by multiplying each column of  $A$  by its column number, then  $\det(B) = 6\det(A)$ .

If the sum of the first and the third row vectors of a  $3 \times 3$  matrix  $A$  is equal to the second row vector of  $A$ , then  $\det(A) = 0$ .

4) Mahesh bought 2 kg potato and  $c$  kg dal from a shop, and paid ₹ 200 to the shopkeeper. Gaurav bought 4 kg potato and 4 kg dal, and paid ₹  $d$  to the shopkeeper. If  $x_1$  represents the price of 1 kg potato and  $x_2$  represents the price of 1 kg dal, then choose the set of correct options. **1 point**

☐ The matrix representation to find  $x_1$  and  $x_2$  is  $\begin{bmatrix} 2 & 4 \\ 4 & c \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 200 \\ d \end{bmatrix}$

☐ The matrix representation to find  $x_1$  and  $x_2$  is  $\begin{bmatrix} 2 & c \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 200 \\ d \end{bmatrix}$

If Mahesh tries to find the price of 1 kg potato and 1 kg dal using appropriate matrix representation by taking  $c = 2$  and  $d = 400$ , then the price of 1 kg potato that he thus arrives at, will not be unique.

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If Mahesh tries to find the price of 1 kg potato and 1 kg dal using the appropriate matrix representation by taking  $c = 2$  and  $d = 400$ , then he will be able to find the price (as a numerical value) of 1 kg potato.

If Mahesh tries to find the price of 1 kg potato and 1 kg dal using the appropriate matrix representation by taking  $c = 2$  and  $d = 400$ , then he will fail to find the price (as a numerical value) of 1 kg potato.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The matrix representation to find  $x_1$  and  $x_2$  is  $\begin{bmatrix} 2 & c \\ 4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 200 \\ d \end{bmatrix}$

If Mahesh tries to find the price of 1 kg potato and 1 kg dal using appropriate matrix representation by taking  $c = 2$  and  $d = 400$ , then the price of 1 kg potato that he thus arrives at, will not be unique.

If Mahesh tries to find the price of 1 kg potato and 1 kg dal using the appropriate matrix representation by taking  $c = 2$  and  $d = 400$ , then he will fail to find the price (as a numerical value) of 1 kg potato.

5) The marks obtained by Safina, Ram and Pratiksha in Quiz 1, Quiz 2 and End sem (with the maximum marks for each exam being 100) are shown in Table W1PT2. **1 point**

|           | Quiz 1 | Quiz 2 | End sem |
|-----------|--------|--------|---------|
| Safina    | 89     | 95     | 88      |
| Ram       | 92     | 81     | 98      |
| Pratiksha | 85     | 93     | 98      |

Table: W1PT2

The weightage of marks in final grade(in percent) of Quiz 1, Quiz 2, and End sem is shown in Table W1PT3.

|         | In percent (%) |
|---------|----------------|
| Quiz 1  | 20             |
| Quiz 2  | 20             |
| End sem | 60             |

Table: W1PT3

Choose the set of correct options.

Final grades (in 100) of Safina, Ram and Pratiksha can be represented by the matrix:

☐

$$\begin{bmatrix} 93.4 \\ 94.4 \\ 89.6 \end{bmatrix}$$

Final grades (in 100) of Safina, Ram and Pratiksha can be represented by the matrix :

☐

$$\begin{bmatrix} 89.6 \\ 93.4 \\ 94.4 \end{bmatrix}$$

- ☐ The order of the matrix which represents final grades(in 100) of Safina, Ram and Pratiksha is  $3 \times 1$

If bonus marks given to Safina, Ram and Pratiksha are represented by the

following matrix  $\begin{bmatrix} 1.4 \\ 2.6 \\ 0 \end{bmatrix}$  then the overall final grades (in 100) can be

- ☐ represented by the matrix:

$$\begin{bmatrix} 91 \\ 96 \\ 94.4 \end{bmatrix}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

Final grades (in 100) of Safina, Ram and Pratiksha can be represented by the matrix :

$$\begin{bmatrix} 89.6 \\ 93.4 \\ 94.4 \end{bmatrix}$$

The order of the matrix which represents final grades(in 100) of Safina, Ram and Pratiksha is  $3 \times 1$

If bonus marks given to Safina, Ram and Pratiksha are represented by the following matrix

$\begin{bmatrix} 1.4 \\ 2.6 \\ 0 \end{bmatrix}$  then the overall final grades (in 100) can be represented by the matrix:

$$\begin{bmatrix} 91 \\ 96 \\ 94.4 \end{bmatrix}$$

**Numerical Answer Type (NAT):**

6) Let  $A$  be a  $3 \times 3$  matrix with non-zero determinant and  $B$  be a matrix obtained by adding 5 times of first row of  $A$  to the third row of  $A$  and adding 10 times of second row of  $A$  to the first row of  $A$ . What is the value of  $\det(2AB^{-1})$ ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 8

**1 point**

7) If  $A = \begin{bmatrix} 4 & 6 & 10 \\ 1 & 4 & -5 \\ 4 & 1 & 10 \end{bmatrix}$ , then find  $\det(A)$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -150

**1 point**

**Comprehension Type Question:**



**(Use the below information for question 8, 9 and 10)**

A shopkeeper sells three types of clothes- shirts, jeans, and T- shirts- in three different sizes: small, medium, and large. In a week, he sold 1 small, 1 medium and 2 large sized shirts; 2 small,  $c$  medium and 6 large sized jeans, and 1 small, 3 medium and  $c - 5$  large sized T-shirts (where  $c$  is an integer). The price of shirts, jeans, and T-shirts remain the same for different sizes (i.e., small, medium, and large sized shirts have the same price; similarly, small, medium, large sized jeans have the same price; and small, medium, large sized T-shirts have same price). The shopkeeper earned ₹7 , ₹27 and ₹43 (in thousand) in that week, for small, medium, and large sized clothes respectively.

8) Let  $s, j, t$  represents the price (in thousand) of 1 shirt, 1 jeans and 1 T-shirt respectively and we want to find  $s, j, t$  by solving a system of linear equations represented by the matrix form  $Ax = b$ , where  $x = (s, j, t)^T$  and  $b = (7, 27, 43)^T$ . If sum of entries in the first row of  $A$  is  $p$ , sum of entries in the second row of  $A$  is  $q$  and sum of entries in the third row of  $A$  is  $r$  (take the topmost row as the first row), then find the value of  $p + q - r$ .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

**1 point**

9) If  $\det(A) = 122$ , then how many medium sized jeans were sold in the week?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 16

**1 point**

10) If  $A$  is the matrix as above (in question 8) and  $\det(A) = 122$ , then what is the price(in thousand) of a T-shirt, where price (in thousand) of a shirt is 2 and price (in thousand) of a jeans is 1?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

**1 point**

Your score is: 0/10